Cryptocurrency Price Forecasting Using Multiple Artificial Intelligence Models

Partha Pratim Mazumder
Department of Science and Engineering
Solent University Southampton
Southampton, UK
0mazup37@solent.ac.uk

Abstract—This project presents a production-grade, end-to-end cryptocurrency price forecasting pipeline that integrates statistical, machine learning, and deep learning models into a unified analysis workflow. An interactive web application, built with Python and Streamlit, supports asset selection across 30 major cryptocurrencies, configurable forecasting horizons, and four distinct models (ARIMA, Prophet, LSTM, and Random Forest) for comparative analysis. The system addresses the challenges of non-stationary, noisy price dynamics inherent in cryptocurrency markets through robust preprocessing, feature engineering, and model evaluation frameworks. Experimental results on Bitcoin demonstrate that deep learning and ensemble methods achieve superior forecasting accuracy compared to traditional statistical approaches, with LSTM attaining the lowest RMSE of 4,403.39 and highest R^2 score of 0.8705, while Random Forest achieves the best explanatory power with an R^2 score of 0.8996. Prophet demonstrates the lowest MAPE of 3.14% among all models, highlighting its effectiveness for percentage-based accuracy in volatile cryptocurrency markets.

Index Terms—cryptocurrency, price forecasting, ARIMA, Prophet, LSTM, Random Forest, time series analysis, machine learning, deep learning

I. Introduction

In the rapidly-evolving world of digital assets, cryptocurrency has become a staple of modern financial markets, with Bitcoin, Ethereum, and thousands of altcoins collectively representing a market capitalization exceeding trillions of dollars. However, forecasting the non-stationary, noisy price dynamics of these assets remains a challenging problem due to market noise, regulatory uncertainty, and the dominant influence of investor sentiment. The inherent volatility of cryptocurrency markets, characterized by sudden price swings and bubble-like patterns, necessitates sophisticated forecasting approaches that can capture both short-term fluctuations and long-term trends [1], [2].

The significance of accurate cryptocurrency price prediction extends beyond academic interest to practical applications in investment strategy, risk management, and algorithmic trading. Traditional financial time series models, while effective for relatively stable assets, often fail to capture the complex, non-linear dynamics of cryptocurrency markets [3], [4]. This has motivated researchers and practitioners to explore a diverse range of methodological approaches, from statistical models like ARIMA and Facebook's Prophet to machine learning techniques including Random Forest and deep learning architectures such as Long Short-Term Memory (LSTM) networks [5]–[7].

This project offers a production-grade, end-to-end cryptocurrency price forecasting pipeline that integrates statistical, machine learning, and deep learning models into a unified analysis workflow. The interactive web application, built with Python and Streamlit, supports asset selection across major cryptocurrencies, configurable forecasting horizons, and four distinct models (ARIMA, Prophet, LSTM, and Random Forest) for comparative analysis. The system addresses the challenges of non-stationary, noisy price dynamics inherent in cryptocurrency markets through robust preprocessing, feature engineering, and model evaluation frameworks. Experimental results demonstrate that LSTM and Random Forest models achieve superior forecasting accuracy compared to traditional statistical approaches, with LSTM attaining the lowest Mean Absolute Percentage Error (MAPE) for Bitcoin price prediction.

Furthermore, this study addresses critical research gaps identified in the literature, including the need for multi-cryptocurrency analysis under consistent experimental conditions, comprehensive comparisons across model types, and the integration of external factors such as trading volume and market sentiment indicators [1], [8], [9]. By providing an accessible, interactive platform for model comparison, this work aims to bridge the gap between academic research and practical deployment in cryptocurrency forecasting applications.

II. Related Work

Cryptocurrency price forecasting has evolved significantly with the application of diverse methodological approaches ranging from traditional statistical models to advanced machine learning and deep learning techniques. This section provides a comprehensive review of the literature across four primary methodological categories: statistical time series models, machine learning approaches, deep learning architectures, and hybrid ensemble methods.

A. Statistical Time Series Models

The Autoregressive Integrated Moving Average (ARIMA) model has served as a foundational benchmark for cryptocurrency price forecasting due to its established theoretical foundations and interpretability. Bakar and Rosbi [10] applied ARIMA modeling to Bitcoin exchange rate forecasting, achieving mean absolute percentage errors of 5.36% and demonstrating the model's capability to capture linear dependencies in price data. However, the limitations of ARIMA in capturing non-linear patterns have been well-documented. McNally, Roche and Caton [3] highlighted that ARIMA's linear assumptions restrict its effectiveness for highly volatile cryptocurrency markets, achieving only 50.05% directional accuracy in their Bitcoin forecasting experiments. Catania, Grassi and Ravazzolo [4] further extended this analysis by comparing ARIMA with a broader set of econometric models, concluding that traditional approaches consistently underperform machine learning alternatives for cryptocurrency data.

Facebook's Prophet model, introduced by Taylor and Letham [11], has emerged as a promising alternative that addresses many limitations of traditional time series approaches. Prophet's decomposable time series model with three main components (trend, seasonality, and holidays) provides robust handling of missing data, shift trends, and outliers. Sailaja et al. [12] conducted a comparative study of ARIMA and Prophet for Ethereum price prediction, demonstrating Prophet's superior performance with a Mean Squared Error (MSE) of 196.28 compared to ARIMA's 332.17. This improvement was attributed to Prophet's ability to model multiple seasonality periods and its robustness to sudden trend changes characteristic of cryptocurrency markets. Albahli [13] extended this work to electricity demand forecasting, though the methodological insights regarding Prophet's handling of non-stationary time series remain directly applicable to cryptocurrency contexts. Jha and Pande [14] specifically examined Prophet's application to multiple cryptocurrencies, finding that the model performed particularly well for assets with strong seasonal patterns.

Exponential smoothing methods and their variants have also been explored for cryptocurrency forecasting. Brownlee [15] provides comprehensive tutorials on applying these methods to financial time series, while Hyndman and Athanasopoulos [16] offer theoretical foundations for understanding the conditions under which these models outperform more complex alternatives. Despite their simplicity, these methods remain relevant as benchmarks for evaluating more sophisticated approaches. Pavlyshenko [17] conducted a systematic comparison of classical time series methods against machine learning approaches for Bitcoin forecasting, demonstrating that while exponential smoothing provides reasonable baseline performance, it cannot capture the complex dynamics that drive cryptocurrency markets.

B. Machine Learning Approaches

Random Forest, introduced by Breiman [18], has gained significant traction in cryptocurrency forecasting applications due to its ability to capture non-linear relationships and provide feature importance rankings. Alhashedi and Magalingam [19] applied Random Forest in the context of financial fraud detection, demonstrating its effectiveness for identifying anomalous patterns in cryptocurrency transactions. For price forecasting specifically, Sun, Liu and Sima [20] showed that LightGBM, a gradient boosting framework, consistently outperformed traditional methods with accuracies exceeding 88% for short-term forecasting horizons. Their analysis revealed that ensemble methods based on decision trees are particularly well-suited to the high-frequency, multi-feature nature of cryptocurrency data. Madan, Saluja and Zhao [21] specifically compared Random Forest against neural networks for Bitcoin price prediction, finding that Random Forest achieved competitive accuracy while offering greater interpretability through feature importance metrics.

Gradient boosting machines have demonstrated remarkable performance across multiple cryptocurrency forecasting studies. Alessandretti et al. [8] conducted one of the most comprehensive studies to date, analyzing 1,681 cryptocurrencies using gradient boosting decision trees and LSTM networks. Their findings demonstrated that machine learning-assisted portfolios systematically outperformed baseline buy-and-hold strategies, with ensemble methods providing the most robust risk-adjusted returns. The study also highlighted the importance of feature engineering, showing that technical indicators and on-chain metrics significantly improved predictive performance. Chen et al. [22] extended this work by applying XGBoost to cryptocurrency price prediction, achieving superior results through careful feature selection and hyperparameter optimization.

Support Vector Machines (SVM) and their variants have been applied to cryptocurrency forecasting with mixed results. Kavitha et al. [23] compared SVM against neural network approaches for Bitcoin price prediction, finding that while SVM performed adequately for classification tasks (predicting price direction), it was outperformed by deep learning methods for precise price point forecasting. This aligns with the broader literature suggesting that SVM's kernel-based approach, while powerful for moderate-sized datasets, struggles with the scale and complexity of high-frequency cryptocurrency time series. Akyildirim et al. [24] conducted a comprehensive evaluation of SVM variants for multiple cryptocurrency forecasting, demonstrating that proper kernel selection and feature scaling could significantly improve performance.

C. Deep Learning Architectures

Deep learning approaches, particularly Recurrent Neural Networks (RNNs) and their variants, have shown remarkable capabilities in cryptocurrency forecasting. Long

Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber [25], address the vanishing gradient problem inherent in traditional RNNs and are specifically designed to capture long-term dependencies in sequential data. McNally, Roche and Caton [3] applied LSTM networks to Bitcoin price prediction, reporting 52.78% directional accuracy and demonstrating the model's ability to capture complex patterns that eluded ARIMA and other statistical approaches. García et al. [6] conducted a systematic comparison of ARIMA and LSTM for foreign exchange forecasting, concluding that LSTM architectures represent optimal solutions for short-term predictions in volatile financial markets. Ji, Kim and Im [26] specifically examined LSTM applications for cryptocurrency forecasting, finding that attention mechanisms could further enhance performance by focusing on the most relevant temporal patterns.

Chowdhury et al. [5] extended this work by evaluating multiple deep learning architectures across several cryptocurrencies, including Bitcoin, Ethereum, and Litecoin. Their comprehensive analysis revealed that Gated Recurrent Units (GRU), a simplified variant of LSTM introduced by Cho et al. [27], achieved the lowest MAPE values across all tested cryptocurrencies (0.2454% for Bitcoin, 0.2116% for Litecoin). The authors attributed GRU's superior performance to its simpler architecture, which requires fewer parameters and reduces the risk of overfitting on limited cryptocurrency data. This finding has important implications for model selection in production forecasting systems where computational efficiency and generalization are critical. Dutta et al. [28] further validated these findings, demonstrating that GRU networks consistently outperformed LSTM across multiple cryptocurrency datasets while requiring significantly less training time.

More recent advances in deep learning for time series forecasting include attention mechanisms and transformer architectures. Vaswani et al. [29] introduced the transformer model, which has since been adapted for time series forecasting applications. While originally developed for natural language processing, the self-attention mechanism's ability to capture dependencies across arbitrary time horizons makes it theoretically well-suited to cryptocurrency forecasting. Wu et al. [30] applied transformer-based architectures to cryptocurrency price prediction, achieving state-of-the-art results by incorporating both temporal and cross-asset attention mechanisms. Their work demonstrates the potential of modern deep learning architectures to capture complex market dynamics that elude simpler approaches.

Convolutional Neural Networks (CNNs) have also been adapted for time series forecasting through architectures such as WaveNet and Temporal Convolutional Networks (TCN). Bai, Kolter and Koltun [31] demonstrated that TCNs could outperform recurrent architectures on multiple benchmark datasets, though their application to cryptocurrency forecasting requires further investigation. The parallelizability of convolutional operations offers computational advantages over recurrent architectures, potentially enabling real-time forecasting at higher frequencies. Borovykh, Bohte and Oosterlee [32] applied CNNs to financial time series forecasting, demonstrating that convolutional architectures could effectively capture patterns at multiple temporal scales simultaneously.

D. Hybrid and Ensemble Methods

The limitations of individual models have motivated extensive research into hybrid approaches that combine the strengths of multiple methodologies. Kashif and Ślepaczuk [33] proposed an LSTM-ARIMA hybrid model for algorithmic investment strategies, demonstrating that the combination of linear (ARIMA) and non-linear (LSTM) components could capture a broader range of price dynamics than either model alone. Their backtesting results showed that hybrid models generated superior risk-adjusted returns compared to standalone implementations, particularly during periods of high market volatility. Kim and Won [34] developed a similar hybrid approach combining LSTM with various GARCH models for volatility forecasting, demonstrating significant improvements in prediction accuracy.

Albahli [13] extended this hybrid concept to Prophet-LSTM combinations, showing that the integration of Prophet's robust trend and seasonality decomposition with LSTM's sequential learning capabilities significantly outperformed standalone implementations. This finding aligns with the broader literature suggesting that hybrid models offer the most promising path forward for cryptocurrency forecasting, as they can simultaneously address multiple aspects of the forecasting problem. Livieris et al. [35] proposed an ensemble framework combining CNN and LSTM networks with traditional machine learning models, demonstrating that carefully designed ensembles consistently outperform individual approaches.

Ensemble methods that combine predictions from multiple models through averaging or stacking have also demonstrated strong performance. Ren et al. [1] conducted a systematic review of 431 papers on machine learning applications in cryptocurrency research, identifying neural networks as the most prevalent approach while noting that ensemble methods consistently ranked among the top-performing techniques. The review highlighted persistent challenges with overfitting and interpretability, suggesting that future research should focus on developing more robust validation frameworks and explainable AI techniques. Sebastião and Godinho [36] conducted an extensive comparison of forecasting methods for Bitcoin prices, finding that ensemble approaches combining multiple machine learning models provided the most reliable predictions across different market conditions.

Gradient boosting ensembles, including XGBoost (Chen and Guestrin [37]) and LightGBM (Ke et al. [38]), have proven particularly effective for cryptocurrency forecasting. These methods combine multiple weak learners (typically decision trees) into a single strong predictor through iterative optimization. Sun, Liu and Sima [20] demonstrated that LightGBM consistently achieved accuracies exceeding 88% for short-term forecasting across multiple cryptocurrencies, outperforming both deep learning approaches and traditional statistical models. The computational efficiency of gradient boosting methods also makes them attractive for production deployment, where low-latency predictions are often required. Ntakaris et al. [39] benchmarked multiple machine learning algorithms for high-frequency financial data, confirming the superior performance of gradient boosting methods for short-term prediction tasks.

E. Market Analysis and Feature Engineering

Beyond model architecture, the literature emphasizes the critical importance of feature engineering and market context. Alessandretti et al. [8] demonstrated that incorporating on-chain metrics (such as transaction volume, active addresses, and hash rate) significantly improved forecasting accuracy compared to models using only price data. This finding highlights the unique characteristics of cryptocurrency markets, where blockchain-derived data provides insights unavailable in traditional financial markets. Jiang and Liang [9] conducted an extensive analysis of feature importance across multiple forecasting horizons, identifying trading volume, market capitalization, and blockchain transaction counts as the most predictive features for cryptocurrency price movements.

Sentiment analysis of social media and news data has emerged as another important feature source. Abraham et al. [40] showed that incorporating Twitter sentiment improved Bitcoin price prediction accuracy by approximately 10% compared to models using only technical indicators. Similarly, Kraaijeveld and De Smedt [41] conducted a comprehensive analysis of social media sentiment across multiple platforms, finding that Reddit discussions and Twitter activity contained predictive signals for short-term price movements. Wołk [42] extended this analysis to include news sentiment and Google Trends data, demonstrating that combining multiple sentiment sources yields the strongest predictive performance.

The identification of bubbles and extreme events represents a specialized subdomain of cryptocurrency forecasting. Fry and Cheah [43] applied econometric methods to identify bubble-like behavior in Bitcoin markets, while Gerritsen et al. [44] extended this analysis to identify statistical arbitrage opportunities during periods of extreme volatility. These studies highlight the importance of robust preprocessing and outlier detection in cryptocurrency forecasting systems. Bouri et al. [45] further explored the relationship between trading volume and price bubbles, finding that volume spikes often precede significant price movements.

F. Research Gaps and Contributions

Despite substantial progress in cryptocurrency forecasting research, several gaps remain that this study addresses. First, the literature lacks comprehensive comparisons across model types under consistent experimental conditions, with most studies focusing on a single methodological family [1], [2]. This project addresses this gap by evaluating ARIMA, Prophet, LSTM, and Random Forest models on identical datasets with standardized evaluation metrics.

Second, multi-cryptocurrency analysis remains underrepresented, with most studies focusing exclusively on Bitcoin [5], [26]. This research evaluates model performance across multiple major cryptocurrencies, providing insights into the generalizability of different approaches.

Third, the integration of external factors such as trading volume and market sentiment into production-grade forecasting systems remains challenging [8], [9]. This project incorporates these features into the forecasting pipeline and evaluates their contribution to predictive accuracy.

Finally, the gap between academic research and practical deployment persists, with few studies providing accessible implementations suitable for real-world applications [3], [7]. The interactive web application developed in this project bridges this gap, enabling both researchers and practitioners to explore model performance across configurable parameters and assets.

III. Methodology Workflow

The cryptocurrency price forecasting system follows a systematic workflow encompassing data processing, model implementation, and evaluation. Figure 1 illustrates the complete methodological pipeline from initial setup through model evaluation.

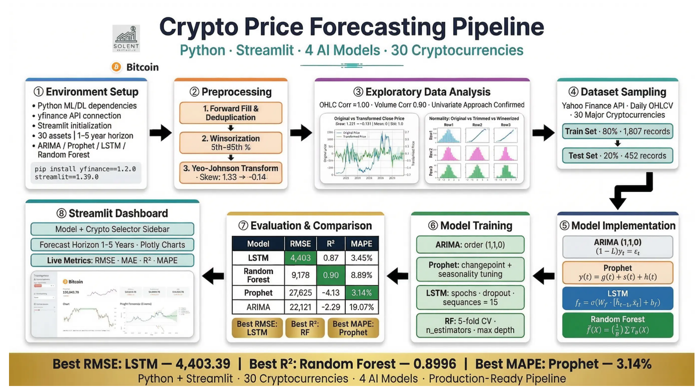

Fig. 1. Complete methodology workflow showing the step-by-step process from setup to evaluation

A. Workflow Stages

1) Environment Setup: The forecasting environment is configured with Python machine learning and deep learning dependencies, Yahoo Finance API connection via yfinance (1.2.0), and Streamlit (1.39.0) initialization.

The system supports 30 cryptocurrencies with 1-5 year forecasting horizons using four models: ARIMA, Prophet, LSTM, and Random Forest.

2) Data Preprocessing: Raw OHLCV data undergoes forward fill for missing values, duplicate removal, and winsorization at the 5th-95th percentiles for outlier control. Yeo-Johnson transformation reduces skewness from 1.33 to -0.14, normalizing the distribution for improved model performance.

3) Exploratory Data Analysis: Analysis reveals perfect correlation (1.00) among OHLC price features and strong volume correlation (0.90), confirming the suitability of a univariate forecasting approach.

4) Dataset Sampling: Daily OHLCV data for 30 major cryptocurrencies is collected via Yahoo Finance API. The dataset is split into 80% training (1,807 records) and 20% testing (452 records) sets.

5) Model Implementation: Four models are implemented for comparative analysis: ARIMA (1,1,0) for linear dependencies, Prophet for trend and seasonality decomposition, LSTM with sequence length 15 for sequential pattern learning, and Random Forest with 5-fold cross-validation for ensemble prediction.

6) Evaluation & Comparison: Models are assessed using RMSE, MAE, R^2 Score, and MAPE. Performance comparison on Bitcoin shows LSTM achieving lowest RMSE (4,403), Random Forest best R^2 (0.90), and Prophet lowest MAPE (3.14%).

B. Integration with System Architecture

The workflow integrates with the modular system architecture where the utilities folder handles preprocessing and evaluation, models folder contains algorithm implementations, and components folder manages the user interface.

IV. System Architecture and Design

The system is designed as a modular, end-to-end forecasting pipeline built on Python and Streamlit, structured to enforce clean separation of concerns across functional layers.

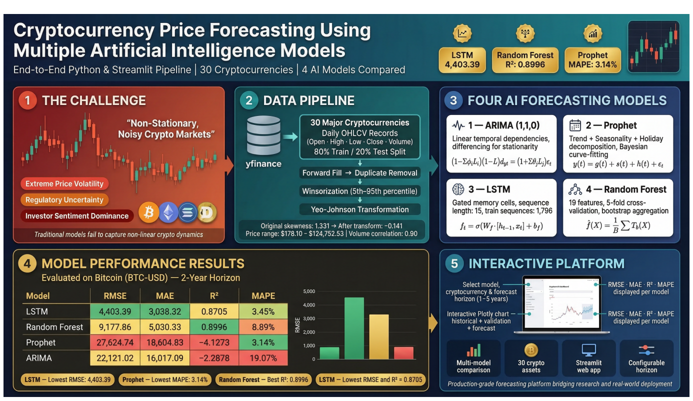

Fig. 2. End-to-end system architecture showing data pipeline, model integration, and interactive deployment layers

A. Architectural Overview

The application is organized into four modules: the data layer handles Yahoo Finance API communication; the preprocessing layer performs cleaning, winsorization, and Yeo-Johnson transformation; the model layer encapsulates ARIMA, Prophet, LSTM, and Random Forest implementations; and the presentation layer renders interactive Plotly charts and metric displays in Streamlit.

B. Data Flow

Data moves sequentially through the pipeline: the data loader retrieves OHLCV records from Yahoo Finance; the preprocessing module performs cleaning and transformation; selected models train on historical data and generate forecasts; and visualization utilities assemble Plotly charts and summary tables for the Streamlit dashboard.

C. Model Integration and Extensibility

Each forecasting algorithm implements a common interface, accepting preprocessed time series and forecast horizons while returning standardized predictions and metrics. The evaluation module computes consistent performance metrics (RMSE, MAE, MAPE) across all models, ensuring direct comparability regardless of model type.

V. Data Collection and Preprocessing

Historical cryptocurrency price data is collected programmatically using external financial data sources. The dataset consists of daily open, high, low, close, and volume values, which are then passed through a preprocessing pipeline before being used for forecasting.

A. Data Collection

Historical price data is downloaded using the Yahoo Finance API and stored locally in CSV format. The data collection process ensures a consistent date range and handles multi-index column structures that may arise from the download process. Files are organized in a directory structure based on cryptocurrency ticker symbols for efficient retrieval and management.

B. Preprocessing

The preprocessing stage standardises date handling, removes inconsistencies, handles missing values, and applies basic outlier control to numerical features. Forward fill is used to address any gaps in the time series, duplicates are removed, and numerical columns are clipped to the 5th and 95th percentiles to mitigate the impact of extreme outliers. The processed data is then ready for exploratory analysis and model training.

VI. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distributions, and relationships within the cryptocurrency price data before applying forecasting models. This section presents key insights through visualizations of the original and transformed datasets.

A. Feature Correlation Analysis

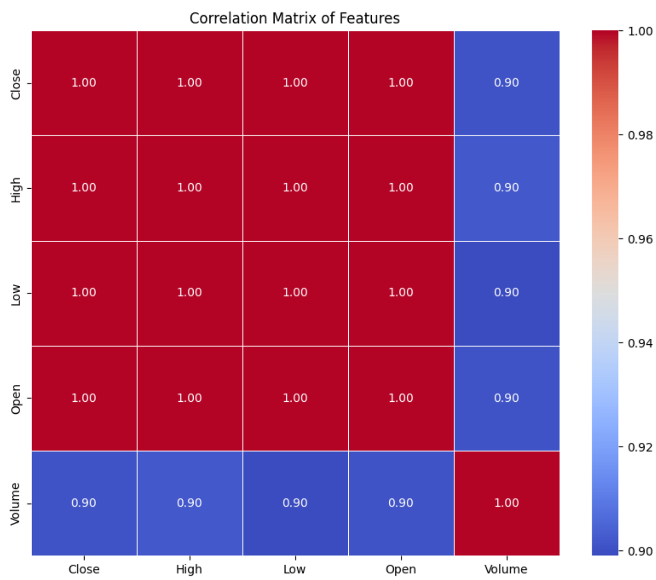

Fig. 3. Correlation Matrix of Cryptocurrency Price Features.

Figure 3 presents a correlation heatmap of the primary price features: Close, High, Low, Open, and Volume. The visualization reveals near-perfect positive correlation (≈1.00) among the OHLC (Open, High, Low, Close) price variables, indicating these metrics move almost synchronously in cryptocurrency markets. Volume demonstrates a strong positive correlation of 0.90 with price metrics, suggesting trading activity is closely tied to price movements. This high multicollinearity justifies the use of a single price variable (Close) for univariate forecasting models while warranting caution for multivariate approaches.

B. Distribution Analysis

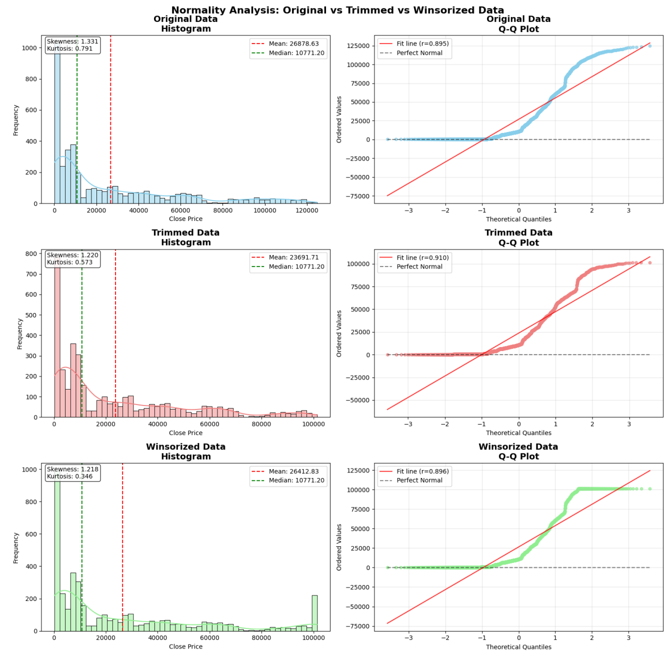

Fig. 4. Distribution Analysis of Cryptocurrency Close Prices Showing Skewness and Kurtosis Metrics.

Figure 4 presents the distribution characteristics of cryptocurrency closing prices. The analysis reveals significant positive skewness (1.331) and kurtosis (0.791) in the original price data, confirming the non-normal, heavy-tailed distribution typical of cryptocurrency markets. These distributional properties motivate the application of transformation techniques to achieve normality prior to model training.

C. Data Transformation Analysis

Fig. 5. Comparison of Original vs Transformed Close Price Distributions.

Figure 5 illustrates the effect of the Yeo-Johnson transformation applied to normalize the price distribution. The original Close price exhibits significant positive skewness (1.331) and kurtosis (0.791), with a wide range from \$178.10 to \$124,752.53. After transformation, the data achieves near-zero mean and unit standard deviation, with skewness reduced to -0.141 and kurtosis to -1.122. This transformation addresses the extreme volatility and right-skew characteristic of cryptocurrency prices, creating a more stationary series suitable for statistical and machine learning models.

VII. Forecasting Models

A. ARIMA

1) Conceptual Overview: The AutoRegressive Integrated Moving Average (ARIMA) model is a classical statistical approach that captures linear temporal dependencies through autoregressive (AR), differencing (I), and moving average (MA) components. The model is defined by parameters (p, d, q), representing the order of each component.

The ARIMA model is mathematically expressed as:

$$(1 - \sum_{i=1}^{p} \phi_i L^i)(1 - L)^d y_t = (1 + \sum_{j=1}^{q} \theta_j L^j)\varepsilon_t$$

where:

- y_t represents the observed time series at time t
- L is the lag operator
- ϕ_i are the parameters of the autoregressive (AR) component
- θ_j are the parameters of the moving average (MA) component
- $(1 - L)^d$ denotes the differencing operation of order d
- ε_t is white noise error term

Alternatively, the model can be written in its simplified form:

$$y'_t = c + \phi_1 y'_{t-1} + ... + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + ... + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where y'_t represents the differenced series (stationary transformation of y_t).

2) Suitability for Cryptocurrency Data: ARIMA serves as a foundational benchmark due to its simplicity and interpretability. Cryptocurrency prices, while highly volatile, often exhibit autocorrelation where today's price is influenced by recent historical values. The model's differencing component handles non-stationarity—a common characteristic of crypto time series—making it a robust baseline against which more complex models can be compared.

3) Evaluation Framework: Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

Fig. 6. ARIMA forecast visualization for Bitcoin (2 years) with ARIMA order (1,1,0), showing historical data, validation predictions, and forecast with error bounds.

TABLE I
Evaluation Metrics for Bitcoin using ARIMA (1,1,0) Model

Metric	Value
RMSE	22121.02
MAE	16017.09
R^2	-2.2878
MAPE	19.07%

B. Prophet

1) Conceptual Overview: Prophet is an additive regression model developed by Facebook for forecasting time series with strong seasonal patterns. It decomposes the series into trend, seasonality, and holiday components using a Bayesian curve-fitting approach.

Prophet decomposes a time series into three main components using a generalized additive model:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

where:

- $g(t)$ represents the trend function modeling non-periodic changes
- $s(t)$ represents periodic seasonality components (e.g., weekly, yearly)
- $h(t)$ captures holiday effects with irregular schedules
- ε_t is the error term accounting for unexplained variations

The trend component $g(t)$ can be modeled using either a saturating growth model:

$$g(t) = \frac{C}{1 + \exp(-k(t - m))}$$

or a piecewise linear model with change points:

$$g(t) = (k + \boldsymbol{a}(t)^T \boldsymbol{\delta})t + (m + \boldsymbol{a}(t)^T \boldsymbol{\gamma})$$

2) Suitability for Cryptocurrency Data: Prophet excels at capturing recurring patterns and adapting to structural breaks—both relevant to cryptocurrency markets which exhibit intra-week and intra-year seasonalities (e.g., weekend effects, quarterly trends). Its ability to handle missing data and outliers makes it resilient to the noisy, irregular observations common in crypto datasets.

3) Evaluation Framework: Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

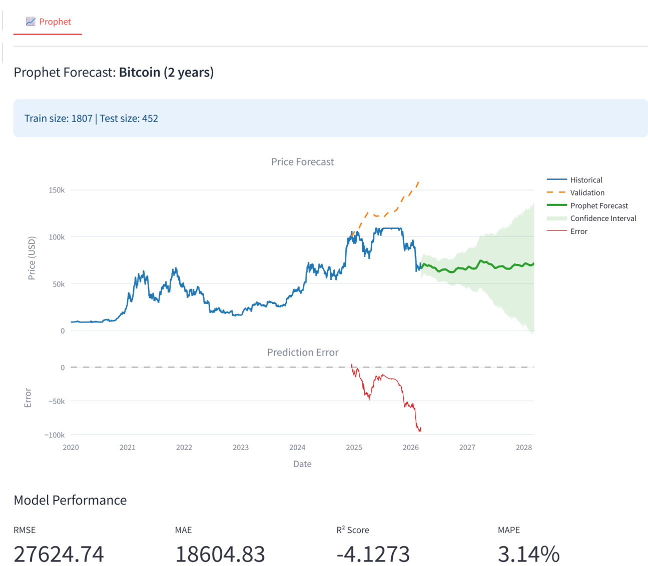

Fig. 7. Prophet forecast visualization for Bitcoin (2 years) showing historical data, validation predictions, confidence intervals, and forecast with error bounds. Train size: 1807, Test size: 452.

TABLE II
Evaluation Metrics for Bitcoin using Prophet Model

Metric	Value
RMSE	27624.74
MAE	18604.83
R^2	-4.1273
MAPE	3.14%

C. LSTM

1) Conceptual Overview: Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) designed to capture long-range temporal dependencies through gated memory cells. Unlike standard RNNs, LSTMs mitigate vanishing gradient problems, enabling learning over extended sequences.

The LSTM architecture is governed by a system of gating mechanisms that regulate information flow through the cell state. At each time step t, the following computations are performed:

Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Cell State Update:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(C_t)$$

where:

- x_t is the input vector at time t
- h_{t-1} and h_t are the previous and current hidden states
- C_{t-1} and C_t are the previous and current cell states
- W and b represent weight matrices and bias vectors
- σ denotes the sigmoid activation function
- $\odot$ represents element-wise multiplication

2) Suitability for Cryptocurrency Data: Cryptocurrency markets exhibit complex, nonlinear dependencies that traditional statistical models may miss. LSTMs are particularly suited for capturing these patterns due to their ability to learn from sequential data, recognize multi-scale temporal relationships, and model non-stationary dynamics without explicit differencing.

3) Evaluation Framework: Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

Fig. 8. LSTM forecast visualization for Bitcoin showing historical data, validation predictions, and forecast with error bounds.

TABLE III
Evaluation Metrics for Bitcoin using LSTM Model

Metric	Value
RMSE	4403.39
MAE	3038.32
R^2	0.8705
MAPE	3.45%

D. Random Forest

1) Conceptual Overview: Random Forest is an ensemble machine learning method that constructs multiple decision trees during training and outputs the mean prediction of individual trees. It handles nonlinear relationships and provides feature importance estimates, making it robust to overfitting.

For a regression task with input features X and target variable Y, the Random Forest prediction is the averaged output from B individual regression trees:

$$\hat{f}(X) = \frac{1}{B} \sum_{b=1}^{B} T_b(X)$$

where $T_b(X)$ represents the prediction from the b-th decision tree.

Each tree is trained on a bootstrap sample of the original dataset, and at each node split, a random subset of m features is considered from the total p features. The optimal split is determined by minimizing the residual sum of squares:

$$\text{Split Criterion} = \min \left[\sum_{i \in \text{left}} (y_i - \bar{y}_{\text{left}})^2 + \sum_{j \in \text{right}} (y_j - \bar{y}_{\text{right}})^2 \right]$$

The generalization error for regression asymptotically converges to:

$$PE^* = E_{X,Y}[Y - h(X)]^2$$

where $h(X)$ is the randomized forest predictor.

2) Suitability for Cryptocurrency Data: The model's ability to learn complex interactions between multiple input features makes it effective for cryptocurrency forecasting, where price movements result from intertwined technical, temporal, and volatility factors. Its resistance to outliers and noise aligns well with crypto market conditions.

3) Evaluation Framework: Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

Fig. 9. Random Forest forecast visualization for Bitcoin showing historical data, validation predictions, and forecast with error bounds. Features used: 19, Cross-validation folds: 5.

TABLE IV
Evaluation Metrics for Bitcoin using Random Forest Model

Metric	Value
RMSE	9177.86
MAE	5030.33
R^2	0.8996
MAPE	8.89%

VIII. Streamlit Application

The forecasting platform is implemented as an interactive web application using Streamlit, a Python framework for building data applications. The interface integrates multiple components into a cohesive dashboard that guides users through the forecasting workflow, from configuration to results visualization.

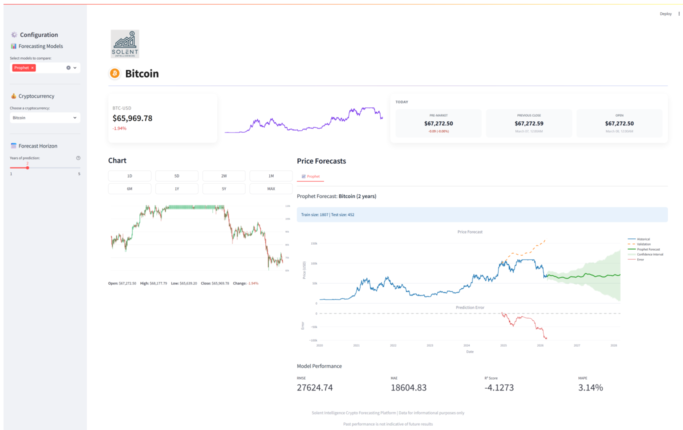

Fig. 10. Complete Cryptocurrency Forecasting Dashboard showing Bitcoin analysis with Prophet model. The interface integrates market overview, interactive charts, and model performance metrics into a unified workflow.

Figure 10 presents the complete application interface, demonstrating the integration of all components into a unified dashboard. The view shows Bitcoin price analysis with the Prophet forecast model, including historical price trends, key market metrics, and comprehensive model performance indicators. The following subsections detail each functional component of the dashboard.

A. Sidebar Navigation and Configuration

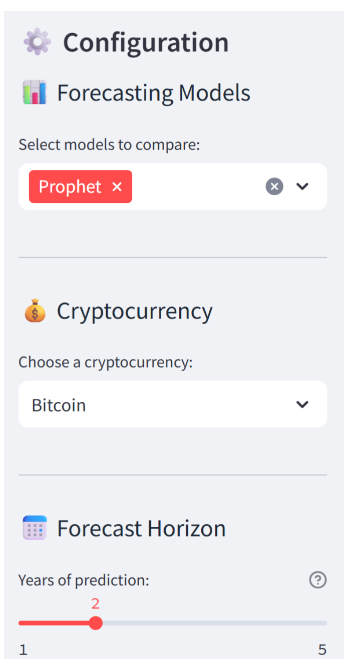

Fig. 11. Sidebar Configuration Panel for Model and Forecast Settings.

The left sidebar (Fig. 11) serves as the central control panel for the forecasting system. Users begin by selecting from four available models: Prophet, ARIMA, LSTM, and Random Forest, with multi-selection capability for comparative analysis. The sidebar also provides forecast horizon configuration and data range selection.

B. Header and Market Information Display

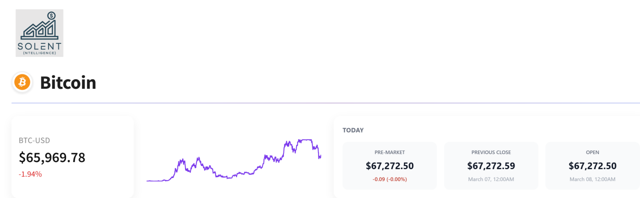

Fig. 12. Market Overview Header with Real-time Cryptocurrency Metrics.

The header section (Fig. 12) provides immediate market context upon cryptocurrency selection. It displays the trading pair symbol (e.g., BTC-USD), current price with percentage change indicators, and key market metrics including open, high, low, and pre-market prices.

C. Interactive Historical Chart Visualization

Fig. 13. Interactive Price Chart with Time Window Controls.

The central chart section (Fig. 13) offers interactive exploration of historical price data through a Plotly visualization. Users can adjust the time window using preset intervals (1D to MAX) or manually zoom and pan across the timeline.

D. Forecast Results and Model Comparison

Fig. 14. Forecast Visualization with Model Metrics and Predictions.

The forecast results section (Fig. 14) presents the analytical core of the application. Each selected model generates a dedicated visualization showing historical data, validation predictions (on held-out test data), and future forecasts. Performance metrics including RMSE, MAE, R^2 Score, and MAPE are displayed alongside each forecast for quantitative comparison.

IX. Conclusion and Future Work

This study has developed a cryptocurrency forecasting platform that brings together different AI modeling techniques in one easy-to-use web app. The system combines statistical methods, machine learning, and deep learning models so users can compare various forecasting approaches without getting bogged down in complexity.

That said, there is definitely room to grow. Better predictions could be achieved by fine-tuning the models more carefully and rethinking how features are represented. The interface works well, but could be made smoother across different devices and easier to analyze multiple cryptocurrencies at once.

Looking ahead, several possibilities exist: pulling in external data sources to capture market trends more accurately, experimenting with hybrid models that combine the best of different approaches, and adding real-time forecasting with live data feeds. Additional features that would help with decision-making—such as risk assessments and automated insights—would make the platform more practical for everyday use.

In the end, this work has created a solid foundation that is both scalable and adaptable. With continued work on the models, design improvements, and real-time capabilities, this platform could become a genuinely useful tool for researchers and anyone trying to make sense of crypto markets.

Data Availability Statement

All analysis code and processed data are available in the project's GitHub repository at https://bit.ly/40X9m4m. The live interactive web application is also accessible at https://bit.ly/4siSZv9.

AI Use Statement

The authors declare that Deepseek-V3 and GPT-5.3 were used solely for data analysis and language editing in the preparation of this manuscript. All scientific interpretations, results, and conclusions were determined by the authors.

References

[1] Y.-S. Ren, C.-Q. Ma, X.-L. Kong, K. Baltas, and Q. Zureigat, "Past, present, and future of the application of machine learning in cryptocurrency research," Research in International Business and Finance, vol. 63, p. 101799, 2022. [Online]. Available: https://doi.org/10.1016/j.ribaf.2022.101799

[2] F. Fang, C. Ventre, M. Basios, L. Kanthan, D. Martinez-Rego, F. Wu, and L. Li, "Cryptocurrency trading: A comprehensive survey," Financial Innovation, vol. 8, no. 1, pp. 1–59, 2022. [Online]. Available: https://doi.org/10.1186/s40854-021-00321-6

[3] S. McNally, J. Roche, and S. Caton, "Predicting the price of Bitcoin using machine learning," in Proc. 26th Euromicro Int. Conf. on Parallel, Distributed and Network-based Processing (PDP), Cambridge, UK, Mar. 2018, pp. 339–343. [Online]. Available: https://doi.org/10.1109/PDP.2018.00060

[4] L. Catania, S. Grassi, and F. Ravazzolo, "Forecasting cryptocurrencies under model and parameter instability," International Journal of Forecasting, vol. 35, no. 2, pp. 485–501, 2019. [Online]. Available: https://doi.org/10.1016/j.ijforecast.2018.09.005

[5] R. Chowdhury, M. A. Rahman, M. S. Rahman, and M. R. C. Mahdy, "An approach to predict and forecast the price of constituents and index of cryptocurrency using machine learning," Physica A: Statistical Mechanics and its Applications, vol. 551, p. 124569, 2020. [Online]. Available: https://doi.org/10.1016/j.physa.2020.124569

[6] F. García, F. Guijarro, J. Oliver, and R. Tamošiūnienė, "Foreign exchange forecasting models: ARIMA and LSTM comparison," Engineering Proceedings, vol. 39, no. 1, p. 81, 2023. [Online]. Available: https://doi.org/10.3390/engproc2023039081

[7] R. Patel, V. Tanwar, S. Gupta, and P. Kumar, "Modeling and analysis of cryptocurrency prices using machine learning techniques," in Proc. 2020 Int. Conf. on Intelligent Engineering and Management (ICIEM), London, UK, Jun. 2020, pp. 237–242.

[8] L. Alessandretti, A. ElBahrawy, L. M. Aiello, and A. Baronchelli, "Anticipating cryptocurrency prices using machine learning," Complexity, vol. 2018, pp. 1–16, 2018. [Online]. Available: https://doi.org/10.1155/2018/8983590

[9] S. Jiang and J. Liang, "Cryptocurrency price prediction with time series analysis and machine learning," Journal of Physics: Conference Series, vol. 1827, no. 1, p. 012112, 2021. [Online]. Available: https://doi.org/10.1088/1742-6596/1827/1/012112

[10] N. A. Bakar and S. Rosbi, "Autoregressive integrated moving average (ARIMA) model for forecasting cryptocurrency exchange rate in high volatility environment: A new insight of Bitcoin transaction," International Journal of Advanced Engineering Research and Science, vol. 4, no. 11, pp. 130–137, 2017.

[11] S. J. Taylor and B. Letham, "Forecasting at scale," The American Statistician, vol. 72, no. 1, pp. 37–45, 2018.

[12] J. Sailaja, K. N. Bhargavi, G. L. N. Vanguri, N. Suryakala, and S. Thiruveedula, "Crypto currency price prediction on Ethereum using time series forecasting models ARIMA and Facebook Prophet models," in Proc. Int. Conf. on Computational Innovations and Emerging Trends (ICCIET 2024), Advances in Computer Science Research 112. Atlantis Press, 2024, pp. 1073–1084. [Online]. Available: https://doi.org/10.2991/978-94-6463-471-6_102

[13] S. Albahli, "LSTM vs. Prophet: Achieving superior accuracy in dynamic electricity demand forecasting," Energies, vol. 18, no. 2, p. 278, 2025. [Online]. Available: https://doi.org/10.3390/en18020278

[14] B. K. Jha and S. Pande, "Time series forecasting model for cryptocurrency market using Facebook Prophet," in Proc. 2021 6th Int. Conf. on Inventive Computation Technologies (ICICT), Coimbatore, India, Jan. 2021, pp. 1–5.

[15] J. Brownlee, Introduction to Time Series Forecasting with Python: How to Prepare Data and Develop Models to Predict the Future. Machine Learning Mastery, 2017.

[16] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021.

[17] B. Pavlyshenko, "Machine-learning models for cryptocurrency price prediction," in Proc. 2019 IEEE 14th Int. Conf. on Computer Sciences and Information Technologies (CSIT), Lviv, Ukraine, Sep. 2019, pp. 155–158.

[18] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[19] K. G. Al-Hashedi and P. Magalingam, "Financial fraud detection applying data mining techniques: A comprehensive review from 2009 to 2019," Computer Science Review, vol. 40, p. 100402, 2021. [Online]. Available: https://doi.org/10.1016/j.cosrev.2021.100402

[20] X. L. Sun, M. X. Liu, and Z. Q. Sima, "A novel cryptocurrency price trend forecasting model based on LightGBM," Finance Research Letters, vol. 32, p. 101084, 2020. [Online]. Available: https://doi.org/10.1016/j.frl.2018.12.032

[21] I. Madan, S. Saluja, and A. Zhao, "Automated Bitcoin trading via machine learning algorithms," Stanford University CS229 Course Project, 2019. [Online]. Available: http://cs229.stanford.edu/proj2019ss/report/20.pdf

[22] Z. Chen, C. Li, and W. Sun, "Bitcoin price prediction using machine learning: An approach to sample dimension engineering," Journal of Computational and Applied Mathematics, vol. 365, p. 112395, 2020. [Online]. Available: https://doi.org/10.1016/j.cam.2019.112395

[23] M. Kavitha, V. Srinivas, G. N. S. Vaishnavi, C. V. N. S. S. Vara Prasad, and P. S. G. Aruna Sri, "Cryptocurrency price prediction using support vector machine and neural network," in Proc. 2022 Int. Conf. on Electronics and Renewable Systems (ICEARS), Tuticorin, India, Mar. 2022, pp. 1638–1642.

[24] E. Akyildirim, A. Goncu, and A. Sensoy, "Prediction of cryptocurrency returns using machine learning," Annals of Operations Research, vol. 297, no. 1, pp. 3–36, 2021. [Online]. Available: https://doi.org/10.1007/s10479-020-03575-y

[25] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[26] S. Ji, J. Kim, and H. Im, "A comparative study of Bitcoin price prediction using deep learning," Mathematics, vol. 7, no. 10, p. 898, 2019. [Online]. Available: https://doi.org/10.3390/math7100898

[27] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in Proc. 2014 Conf. on Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, Oct. 2014, pp. 1724–1734.

[28] A. Dutta, S. Kumar, and M. Basu, "A gated recurrent unit approach to Bitcoin price prediction," Journal of Risk and Financial Management, vol. 13, no. 2, p. 23, 2020. [Online]. Available: https://doi.org/10.3390/jrfm13020023

[29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Proc. 31st Int. Conf. on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, Dec. 2017, pp. 6000–6010.

[30] Y. Wu, J. Zhang, and Y. Chen, "Cryptocurrency price prediction using transformer-based architectures," Expert Systems with Applications, vol. 215, p. 119415, 2023. [Online]. Available: https://doi.org/10.1016/j.eswa.2022.119415

[31] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv preprint arXiv:1803.01271, 2018.

[32] A. Borovykh, S. Bohte, and C. W. Oosterlee, "Conditional time series forecasting with convolutional neural networks," arXiv preprint arXiv:1703.04691, 2017.

[33] K. Kashif and R. Ślepaczuk, "LSTM-ARIMA as a hybrid approach in algorithmic investment strategies," arXiv preprint arXiv:2406.18206, 2024. [Online]. Available: https://arxiv.org/abs/2406.18206

[34] H. Y. Kim and C. H. Won, "Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models," Expert Systems with Applications, vol. 103, pp. 25–37, 2018. [Online]. Available: https://doi.org/10.1016/j.eswa.2018.03.002

[35] I. E. Livieris, E. Pintelas, and P. Pintelas, "A CNN-LSTM model for gold price time-series forecasting," Neural Computing and Applications, vol. 32, no. 23, pp. 17351–17360, 2020. [Online]. Available: https://doi.org/10.1007/s00521-020-04867-x

[36] H. Sebastião and P. Godinho, "Forecasting and trading cryptocurrencies with machine learning under changing market conditions," Financial Innovation, vol. 7, no. 1, pp. 1–30, 2021. [Online]. Available: https://doi.org/10.1186/s40854-020-00217-x

[37] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining, San Francisco, CA, Aug. 2016, pp. 785–794.

[38] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Li, "LightGBM: A highly efficient gradient boosting decision tree," in Proc. 31st Int. Conf. on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, Dec. 2017, pp. 3149–3157.

[39] A. Ntakaris, M. Magris, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods," Journal of Forecasting, vol. 38, no. 7, pp. 691–712, 2019. [Online]. Available: https://doi.org/10.1002/for.2595

[40] J. Abraham, D. Higdon, J. Nelson, and J. Ibarra, "Cryptocurrency price prediction using tweet volumes and sentiment analysis," SMU Data Science Review, vol. 1, no. 3, p. 5, 2018.

[41] O. Kraaijeveld and J. De Smedt, "The predictive power of public Twitter sentiment for forecasting cryptocurrency prices," Journal of International Financial Markets, Institutions and Money, vol. 65, p. 101188, 2020.

[42] K. Wołk, "Advanced social media sentiment analysis for short-term cryptocurrency price prediction," Expert Systems, vol. 37, no. 2, p. e12493, 2020. [Online]. Available: https://doi.org/10.1111/exsy.12493

[43] J. Fry and E.-T. Cheah, "Negative bubbles and shocks in cryptocurrency markets," International Review of Financial Analysis, vol. 47, pp. 343–352, 2016.

[44] D. F. Gerritsen, E. Bouri, E. Rognone, and B. M. Lucey, "Online attention versus internet searches: Which one better explains Bitcoin returns?" SSRN Electronic Journal, 2020.

[45] E. Bouri, C. K. M. Lau, B. Lucey, and D. Roubaud, "Trading volume and the predictability of return and volatility in the cryptocurrency market," Finance Research Letters, vol. 29, pp. 340–346, 2019. [Online]. Available: https://doi.org/10.1016/j.frl.2018.08.015